Software Engineering Project Jan-2026

AI Tools Usage & Prompting Report for Milestone 1

Milestone 1 Deliverables

Submitted By

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Team Code : Team-003

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Small Business Operations Platform for Rural Logistics Management

Project Details

Software Name: Cargo-Core

Department Context: Local-Logistic-Department

System Type: Small Business Logistics & Resource Management Platform

Date of Submission : 21-02-2026

Team Details

Team Name: QuadCore-Devs
Team Code: Team-003

Name	Roll Number	Role
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1. Overview of AI Integration

In accordance with the project guidelines, our team actively utilized Generative AI tools (ChatGPT/Gemini) during Milestone 1 to assist with requirement elicitation and data structuring. The AI was strictly used as a collaborative assistant to format raw data into academic Agile standards and to visualize software concepts.

Primary Tools Used:

- Generative AI (ChatGPT-4 / Google Gemini)

2. Areas of Application & Prompt Engineering

Application A: Pain Point Extraction from Raw Transcripts

Objective: To objectively analyze our translated Kannada/Hindi interview transcripts and identify systemic business bottlenecks without human bias.

- **Our Prompt:** > *"Act as an Agile Business Analyst. Read the following interview transcript from a local agricultural logistics Dispatcher. Extract the top 3 operational pain points, focusing on resource allocation, routing, and time management. Transcript: [Inserted the Interviews]."*
- **AI Output:** Identified Resource Mismatch (assigning 4 labourers to a 2-seater vehicle), Inefficient Routing (wasting diesel bouncing between villages), and Lack of Service Time Blocking.
- **Team Refinement:** We verified these points against our field notes and adopted them as the core justifications for our **Multi-Stop Route Optimization** and **Capacity Validation** features.

Application B: Formatting Agile User Stories

Objective: To convert our identified pain points into strict, industry-standard SMART User Stories.

- **Our Prompt:** > *"Convert this pain point: 'The warehouse manager loses expensive plastic crates because drivers forget them at farms' into a standard Agile User Story using the format: 'As a [Role], I want to [Action], so that [Benefit]'. Categorize it under an Epic called 'Inventory Management'."*
- **AI Output:** *"As a Warehouse Manager, I want to digitally track which driver checked out returnable plastic crates, so that I can hold them accountable and reduce financial loss from missing assets."*
- **Team Refinement:** We grouped the AI-formatted stories by Role (Warehouse Manager, Dispatcher, Customer, etc.) to ensure a consistent design hierarchy for our **Milestone_1_Deliverables** document. We also manually ensured that bots were not listed as primary actors (e.g., changing "As an AI Bot" to "As a Customer interacting with an AI").

Application C: Ideation of Software Solutions

Objective: To brainstorm feasible SaaS features that solve rural logistics problems while fitting within a "Small Business" (10-50 employees) budget and scope.

- **Our Prompt:** > *"A small logistics godown struggles with 'Search Fatigue' because fertilizers and food grains are mixed up, and labourers walk miles searching for items. Suggest 3 software features to solve this."*

- **AI Output:** Suggested 1) RFID Tagging, 2) Digital Twin 2D Mapping, and 3) Voice-directed picking.
- **Team Refinement:** We rejected RFID tagging as too expensive for a small business. We selected **Digital Twin 2D Mapping** (Smart Zones) as it only requires a basic tablet UI for the manager to visually separate hazardous fertilizers from seeds.

Application D: UI/UX Prototyping (Feasibility Check)

Objective: To prove that a "Digital Godown Map" could be easily visualized in a React frontend for our final project phase.

- **Our Prompt:** > *"Write a simple React component using Tailwind CSS that displays a 2D grid map of a warehouse. Use dif erent colors to represent 'Fertilizers' (Red) and 'Seeds' (Green). Add a button that simulates an 'Optimized Pick Path' for a labourer."*
- **AI Output:** Generated a functional WarehouseMap.jsx component with an interactive grid and a simulated shortest-path algorithm.
- **Team Refinement:** We tested this code prototype to validate that rendering a 2D array grid is technically feasible for our frontend developers in upcoming milestones.

3. Academic Integrity Declaration

We declare that all business logic, user personas, and core operational problems described in this milestone are derived from **real-world research and authentic local interviews** conducted by Team 003. Generative AI was solely used for structural formatting, grammar refinement, and code prototyping, ensuring that the final deliverables meet high software engineering standards while reflecting genuine local logistics challenges.